Leak test (ntuple, file 3, 20 events): AUC(unlinked vs linked) gain of the emitted angles over each probe; HGB trained on even events, tested on odd (and swapped)

group	n	frac_unlinked	F1_base	F1_gain	F1_gain_swapped_halves	strong_base	strong_gain	strong_gain_swapped_halves	fast_study_gain_F1	fast_study_gain_strong(own CNN)
IL2-4	363950	0.1287	0.7762	0.009514	0.01107	0.8012	0.0002187	-1.928e-05	0.0108	-0.0001
IL1	180648	0.1057	0.8017	0.0009529	0.001411	0.8204	0.001581	-0.0008817	0.0051	0.0022
ID1-4	910220	0.1829	0.7518	-0.0005538	-0.0004332	0.7541	3.428e-05	-0.0001272	0.0001	-0.0001

Leak test (nano, file 3): gain of ALL emu angle columns (cot, sigma, has, sat) over the nano probe

group	auc_no_angles	auc_with_all_angle_columns	gain	n
IL2-4	0.7815	0.7905	0.008956	1815061
IL1	0.8167	0.8213	0.004675	899091
ID1-4	0.7527	0.7525	-0.0001571	4532754

hasAlpha / hasBeta, linked (tpKey >= 0) vs unlinked (must be exactly 1)

group	linked	hasAlpha	hasBeta	n
IL2-4	True	1	1	1583251
IL2-4	False	1	1	231810
IL1	True	1	1	803522
IL1	False	1	1	95569
ID1-4	True	1	1	3706158
ID1-4	False	1	1	826596

Class fractions (TRUTH-ONLY tpEmuClass) vs the prototype (no pT cut)

group	fracT	prototype	fracC_linked	fracC_unlinked	n
IL2-4	0.6468	0.64	0.2254	0.1277	1815061
IL1	0.8007	0.8	0.09299	0.1063	899091
ID1-4	0.7125	0.71	0.1052	0.1824	4532754